

Introduction

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This document provides context for some of the concepts and analyses that we are implementing and planning for the caribou herds overlapping the Kaska Dene Traditional Territory (KDTT). Many of the sections have been generated using Gemini and, in some cases, modified by the BEACONS Project. The sections are added as they are needed and, as such, are not yet structured into a comprehensive well flowing document.

Defining seasonal migration periods

- NSD

Estimating annual and seasonal ranges

- Isopleth levels

Identifying seasonal migration pathways

- Migration mapping

Isopleth levels

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Isopleth Levels

1. Introduction

The concept of an animal's home range, defined as the area traversed by an individual during its normal activities of food gathering, mating, and caring for young, forms a cornerstone of ecological research. Understanding the spatial extent and patterns of animal movement is crucial for addressing a wide array of ecological questions, including habitat use, social interactions, population dynamics, and conservation strategies. In an era marked by increasing human impact on natural environments, the ability to accurately delineate and analyze home ranges is more critical than ever, informing management decisions aimed at mitigating human-wildlife conflicts and ensuring the persistence of biodiversity.

While the basic idea of a home range is intuitive, its quantification presents considerable challenges. Animals rarely utilize their environment uniformly; instead, they exhibit varying intensities of space use across their range. Isopleths, defined as lines connecting points of equal value, have emerged as a powerful tool in spatial ecology to address this complexity. In the context of home range estimation, isopleths provide a probabilistic framework for understanding the likelihood of an animal's presence within a given area. This report aims to provide a comprehensive and expert-level analysis of isopleth levels in home range estimation. It will delve into the definition and role of isopleths, the interpretation of different probability levels, the common methods used to generate them, their applications in ecological analysis, and the crucial limitations and considerations that researchers must be aware of when employing this technique.

2. Defining Isopleths in Home Range Estimation

In ecological studies and wildlife research, an isopleth is a contour line drawn on a map that connects points representing an equal probability density of an animal's location. These lines are not arbitrary; they are derived from an underlying statistical representation of an animal's space use known as the utilization distribution (UD). The UD essentially maps the relative frequency with which an animal is found in different parts of its environment over a specific period, thereby quantifying the intensity of space use.

The concept of the UD can be visualized as a three-dimensional surface draped over the landscape, where the height of the surface at any given point corresponds to the probability density of the animal being located there. The total volume under this surface is normalized to sum to 1, representing the total probability of the animal being somewhere within its range. Isopleths are then the two-dimensional contours formed by slicing this three-dimensional probability surface at different height levels, corresponding to different probabilities of occurrence. These contours delineate areas within which there is a specified probability of finding the animal.

This probabilistic approach offered by isopleths stands in stark contrast to earlier, more simplistic methods of home range estimation, such as the Minimum Convex Polygon (MCP). While the MCP method simply connects the outermost recorded locations of an animal to form a polygon, it provides no information about how the animal uses the space within that boundary. Isopleths, by being derived from the UD, offer a much more nuanced understanding of space use, revealing areas of high and low intensity of activity and allowing for a probabilistic interpretation of an animal's spatial behavior.

3. The Role of Isopleths in Delineating Home Ranges

Isopleths serve as the operational tool for defining the spatial extent of an animal's home range by

allowing researchers to select a specific probability level that corresponds to their definition of the range. For instance, a researcher might choose the 95% isopleth to represent the overall home range, implying that there is a 95% probability of finding the animal within the area enclosed by this contour at any given time.

Different isopleth levels can delineate distinct zones of space use intensity within the broader home range. For example, the 50% isopleth is commonly used to identify the core area, representing the region where the animal spends the majority of its time or has the highest probability of being located. Conversely, higher percentage isopleths encompass progressively larger areas that include the less frequently visited parts of the home range.

The selection of the isopleth level used to define the home range boundary is a critical decision that can significantly influence the estimated size and shape of the range. This choice should not be arbitrary but rather should be guided by the specific ecological question being addressed and the biological characteristics of the species under investigation. For example, a study focusing on resource acquisition might prioritize the core area defined by a lower isopleth, while a study on dispersal patterns might be more interested in the overall extent defined by a higher isopleth.

Furthermore, isopleths provide a standardized and quantitative way to compare home range characteristics between individuals, populations, or across different temporal scales. By comparing the areas enclosed by isopleths at the same probability level (e.g., 95%) for different animals, researchers can assess differences in home range size. Similarly, analyzing the degree of overlap between isopleths of different individuals can provide insights into social interactions and territoriality.

4. Understanding Different Isopleth Levels and Their Probabilistic Interpretation

Different isopleth levels represent different probabilities of an animal's location within its home range. Understanding these probabilities is crucial for interpreting ecological patterns and processes.

50% Isopleth: The 50% isopleth delineates the area where there is a 50% probability of finding the animal at any random point in time within the study period. Ecologically, this is commonly interpreted as the

core area of an animal's home range, representing the zone of most intensive use. This area likely encompasses key resources, such as primary foraging sites, crucial resting locations, or important areas for social interactions. For instance, comparisons of core area size and location can reveal differences in resource availability or social status among individuals.

90% or 95% Isopleth: These higher-level isopleths, typically the 90% or 95% contours, are generally used to represent the overall home range extent. The 95% isopleth, for example, indicates the area within which there is a 95% probability of finding the animal. These levels capture the total area an animal regularly uses, including both the intensively used core and the more peripheral areas that may be visited less frequently for exploration or accessing seasonally available resources. While 95% is a common standard, some researchers advocate for the use of the 90% isopleth in specific contexts.

Other isopleth levels, such as 75% or 99%, can also be used depending on the specific research objectives. For instance, a 75% isopleth might be used to define a secondary area of high use, while a 99% isopleth could represent the maximum extent of occasional excursions.

It is also important to understand the concept of "percent volume contours" in the context of three-dimensional utilization distributions. A 2D isopleth at a given percentage can be thought of as the projection of the corresponding volume contour from the 3D UD onto a two-dimensional map. This means that the area enclosed by an isopleth represents a specific proportion of the total probability volume of the animal's space use.

Ultimately, the choice of which isopleth levels to analyze and interpret should be driven by the specific ecological questions being asked and the biological characteristics of the study animal. Relying solely on arbitrary conventions like always using the 50% and 95% isopleths without considering their ecological relevance to the species and the study system can lead to misleading interpretations.

Table 1: Common Isopleth Levels and Their Typical Ecological Interpretations

Isopleth Level (%)	Typical Ecological Interpretation	Common Applications
50%	Core Area, Area of Highest Use Intensity	Identifying primary foraging or resting sites, Comparing core area size across individuals or conditions
75%	Secondary Area of High Use	Investigating patterns of resource use beyond the core
90%	Overall Home Range Extent (Alternative)	Estimating the total area regularly used, Assessing habitat connectivity
95%	Overall Home Range Extent (Common Standard)	Defining the boundary of the home range for comparisons, Conservation planning
99%	Maximum Extent of Use, Including Occasional Excursions	Identifying rare or exploratory movements

5. Common Methods for Generating Home Range Isopleths

Several methods are commonly employed to generate isopleths for home range estimation, each with its own underlying assumptions and computational approaches.

Kernel Density Estimation (KDE): Kernel Density Estimation is one of the most widely used non-parametric methods for estimating an animal's utilization distribution from a set of location data. The fundamental principle of KDE involves placing a kernel function, which is a probability density function (often a bivariate normal distribution), centered at each observed location of the animal. These individual kernels are then summed across all locations to produce an overall estimate of the probability density of the animal's presence at any given point in space.

A critical aspect of KDE is the selection of the bandwidth or smoothing parameter. The bandwidth determines the size and shape of the kernel and significantly influences the resulting UD and the derived isopleths. A small bandwidth results in a highly localized and potentially fragmented UD, reflecting fine-scale movements, while a large bandwidth produces a smoother, more generalized UD that may obscure important details. Several methods exist for bandwidth selection, including least-squares cross-validation (LSCV) and plug-in methods, but the choice often involves a degree of subjectivity and can substantially impact the results.

KDE offers several advantages, including its ability to estimate complex, non-linear home range shapes that are not constrained by convexity, and its provision of a continuous utilization distribution from which isopleths at any desired probability level can be extracted. However, KDE also has limitations. It is sensitive to the choice of bandwidth, can potentially overestimate home range size, particularly for animals with linear movement patterns or home ranges containing large areas of unused space. Furthermore, traditional KDE assumes that the location data are statistically independent, an assumption that is often violated by GPS telemetry data which typically exhibit temporal autocorrelation.

Minimum Convex Polygon (MCP): The Minimum Convex Polygon is a more traditional and geometrically straightforward method for home range estimation. It defines the home range as the smallest convex polygon that encloses all the recorded locations of an animal. While a 100% MCP

includes all data points, variations exist, such as the 95% MCP, which excludes a small percentage (typically 5%) of the most outlying locations to reduce the influence of occasional excursions.

While isopleths derived from UD's represent probability densities, MCPs essentially define a boundary based on the outermost occurrences. However, one can conceptually link MCPs to isopleths by considering polygons that enclose a specific percentage of the data points.

The primary advantages of MCP are its simplicity in calculation and interpretation. Its historical prevalence also allows for comparisons with older studies. However, MCP suffers from significant limitations. It tends to overestimate home range size, often including large areas that are rarely or never used by the animal. It is also highly sensitive to the inclusion of even a single outlying location, which can disproportionately inflate the estimated range. Critically, MCP provides no information about the intensity of space use within the defined polygon.

Local Convex Hull (LoCoH): Local Convex Hull is a non-parametric kernel method that offers an alternative to traditional KDE, particularly for home ranges with complex shapes. LoCoH constructs a series of overlapping local convex hulls based on subsets of the animal's location data. These subsets are typically defined by a fixed number of nearest neighbors (k-LoCoH), a fixed radius around each point (r-LoCoH), or an adaptive radius based on the distance to neighboring points (a-LoCoH).

Utilization distributions and isopleths can be derived from the density of these overlapping local hulls. LoCoH offers advantages in its ability to better represent home ranges with intricate boundaries, including those with internal unused areas or "hard" boundaries such as coastlines or rivers. Research suggests that LoCoH may provide more accurate estimates and exhibit better convergence properties as sample size increases compared to parametric KDE methods.

Other Methods: Beyond KDE, MCP, and LoCoH, several other methods exist for home range estimation that can generate utilization distributions and, consequently, isopleths. These include:

- **Brownian Bridge Movement Models (BBMM):** These models explicitly incorporate the animal's movement path and the temporal autocorrelation between consecutive locations, making them well-suited for analyzing GPS tracking data.
- **Autocorrelated Kernel Density Estimation (aKDE):** This is a modification of traditional KDE that accounts for the temporal autocorrelation inherent in many animal tracking datasets, providing more robust estimates for serially correlated locations.
- **Time-Geographic Density Estimation (TGDE):** This approach generates a continuous probability density surface of an animal's movements by considering both the spatial locations and the time elapsed between them, incorporating constraints on the animal's maximum velocity.
- **Density-based Fuzzy Home Range Estimator (DFHRE):** This novel method aims to improve home range estimation by incorporating environmental factors such as terrain and obstacles, which can influence animal movement and space use.

Table 2: Common Methods for Generating Home Range Isopleths

Method	Type	Key Principles	Advantages	Limitations
Kernel Density Estimation (KDE)	Non-parametric	Places a kernel function over each location point and sums them to estimate probability density.	Can estimate complex shapes, provides a continuous utilization distribution.	Sensitive to bandwidth selection, potential for overestimation, assumes data independence.
Minimum Convex Polygon (MCP)	Geometric	Creates the smallest convex polygon enclosing all location points.	Simple to calculate and interpret, historically used.	Overestimates home range size, sensitive to outliers, no information on intensity of use.
Local Convex Hull (LoCoH)	Non-parametric	Constructs overlapping local convex hulls based	Handles complex shapes and	Parameter selection can be influential.

	ic	on subsets of data points.	boundaries well, potentially more accurate than KDE.	
Brownian Bridge Movement Model (BBMM)	Probabilistic	Incorporates movement path and temporal autocorrelation.	Suitable for GPS data, accounts for movement patterns.	Can be computationally intensive.
Autocorrelated Kernel Density Estimation (aKDE)	Non-parametric	Modifies KDE to account for temporal autocorrelation.	More robust for serially correlated data.	Still requires bandwidth selection.

6. Interpretation and Application of Isopleth Levels in Ecological Analysis

Isopleth levels are invaluable tools for interpreting and analyzing animal space use in various ecological contexts.

Identifying Core Areas of Activity: The 50% isopleth is a widely adopted standard for delineating the core area of an animal's home range. This area of highest probability of use often reflects the locations of essential resources or habitats that are critical for the animal's survival and reproduction. Researchers can use the 50% isopleth to identify key foraging sites, important resting locations, or areas crucial for social interactions. Comparing the size, shape, and location of core areas across individuals or under different environmental conditions can provide insights into resource competition, habitat quality, or the impact of disturbances.

Defining the Overall Home Range Extent: Higher isopleth levels, typically 90% or 95%, are used to estimate the total spatial extent of an animal's home range. This provides a measure of the total area an animal regularly utilizes, including both the core and the less frequently visited peripheral regions. Understanding the overall home range extent is crucial for conservation planning, as it helps determine the amount of habitat required to support individuals or populations. It can also be important for studying dispersal patterns and potential interactions with human activities or landscape features.

Comparing Space Use Patterns: Isopleths at different probability levels offer a valuable tool for comparing space use patterns among individuals, between populations, or across different time periods or environmental conditions. For instance, comparing the 50% isopleths of two individuals might reveal differences in their core resource areas, while comparing their 95% isopleths could indicate variations in overall range size. Furthermore, changes in isopleth size or location over time can reflect responses to changes in resource availability, the presence of competitors or predators, or habitat alterations. Overlap indices, calculated using the utilization distributions from which isopleths are derived, provide quantitative measures of the degree of spatial overlap between individuals or groups.

Analyzing Habitat Selection and Resource Utilization: By overlaying isopleth maps with spatial data on habitat types, resource distribution, or other environmental variables, researchers can analyze habitat selection patterns within different zones of space use intensity. For example, comparing the proportion of different habitat types within the 50% isopleth (core area) to that within the 95% isopleth (overall range) can reveal whether an animal exhibits specific habitat preferences in its most intensively used areas. This can provide valuable insights into the factors driving habitat selection and resource use.

Visualizing Intensity of Use: Isopleth maps offer a powerful visual representation of the intensity with which an animal utilizes its space across its home range. The spacing of isopleth lines indicates the gradient of use intensity; closely spaced lines signify a rapid change in the probability of occurrence, while widely spaced lines suggest a more gradual transition. This visual representation can be crucial for communicating research findings and understanding the spatial ecology of a

species.

7. Limitations and Considerations When Using Isopleth Levels

While isopleth levels are a valuable tool in home range analysis, it is crucial to be aware of their limitations and to consider several factors when interpreting and applying them.

The influence of sample size and data collection methods cannot be overstated. The accuracy and reliability of home range estimates, and consequently the derived isopleths, are directly dependent on having a sufficient number of location data points that are representative of the animal's space use over the relevant time period. Insufficient data can lead to underestimation or misrepresentation of the home range. Furthermore, the method of data collection, such as GPS telemetry versus VHF radio telemetry, can introduce biases due to factors like location accuracy, fix rates, and potential for signal obstruction.

Spatial and temporal autocorrelation in animal tracking data pose a significant challenge for many traditional home range estimation methods, including standard Kernel Density Estimation. The non-independence of consecutive location points, where points closer in time are likely to be closer in space, violates the assumption of independence underlying these methods. This can lead to biased estimates of home range size and shape. Researchers should consider using methods specifically designed to address autocorrelation, such as Autocorrelated Kernel Density Estimation (aKDE) or Brownian Bridge Movement Models (BBMM).

The subjectivity in parameter selection for certain home range estimation methods is another important consideration. For instance, the choice of bandwidth in KDE can significantly influence the resulting utilization distribution and isopleths, and different bandwidth selection methods can yield varying outcomes. Similarly, parameters in other methods like LoCoH (e.g., the number of neighbors) can also affect the results. Researchers should be aware of this subjectivity and, where possible, justify their parameter choices based on ecological knowledge or employ sensitivity analyses to assess the impact of different parameter values.

Different home range estimation methods have inherent biases and assumptions that can affect the interpretation of isopleth levels. For example, MCPs are known to overestimate home range size, while traditional KDE might smooth over important fine-scale patterns or extend into areas that are physically inaccessible to the animal. The choice of method should be carefully considered in relation to the specific characteristics of the animal's movement and the research question.

The selection of appropriate isopleth levels is crucial for meaningful ecological interpretation. Relying on arbitrary conventions like always using 50% and 95% isopleths without considering their ecological relevance to the species and the study system can lead to misleading conclusions. The choice should be driven by the specific research question and the biological significance of those levels for the animal under study.

Finally, standard home range estimation methods may not adequately account for environmental barriers and habitat heterogeneity. For example, a traditional KDE might produce isopleths that extend across a large river that the animal cannot cross. Newer methods, such as those incorporating cost distance or network-based approaches, can help address these limitations. Additionally, the heterogeneity of the habitat can influence space use patterns, and researchers should consider how this might be reflected in the shape and size of isopleths.

8. Conclusion

Isopleth levels represent a sophisticated and valuable tool in the field of spatial ecology for understanding how animals utilize their environment. By providing a probabilistic framework for delineating home ranges and visualizing the intensity of space use, isopleths offer significant advantages over simpler methods. They enable researchers to identify core areas of activity, estimate overall home range extent, compare space use patterns across different groups or

conditions, and analyze habitat selection within varying intensity-of-use zones.

However, the effective and responsible application of isopleth levels requires a thorough understanding of the underlying principles of home range estimation methods, as well as a critical awareness of their inherent limitations and assumptions. Factors such as sample size, data collection techniques, spatial and temporal autocorrelation, subjectivity in parameter selection, and the influence of environmental barriers must be carefully considered to ensure the generation of reliable and ecologically meaningful home range estimates. The selection of appropriate isopleth levels should be driven by the specific research questions and the biological characteristics of the species under investigation, rather than relying on arbitrary conventions.

In conclusion, isopleth analysis remains a cornerstone of ecological and wildlife research, providing essential insights into animal space use. By employing these techniques thoughtfully and with a clear understanding of their strengths and weaknesses, researchers can continue to advance our knowledge of animal behavior, population ecology, and the spatial dynamics of species in a changing world.

Net Squared Displacement (NSD)

In ecology, particularly in animal movement studies, an "NSD vs. time of year plot" describes the Net Squared Displacement (NSD) of an individual animal over the course of a year.

Let's break down what that means:

- **Net Squared Displacement (NSD):** This is a measure of how far an animal has moved from its initial starting point.
 - You take the straight-line distance between the animal's first recorded location and each subsequent location.
 - Then, you square that distance.
 - So, at any given time, NSD represents the squared straight-line distance from the very beginning of the tracking period.
- **Time of Year:** This refers to the progression of days, weeks, or months within a calendar year.

What an NSD vs. time of year plot describes:

When you plot NSD on the y-axis against the time of year (or simply time since the start of tracking) on the x-axis, you create a visual representation of an animal's movement patterns throughout the year. This type of plot is particularly useful for:

1. **Identifying Migration:**
 - **Migratory animals:** An NSD plot for a migratory animal will often show a characteristic "double sigmoid" or "S-shaped" curve.
 - It starts low, then increases as the animal moves to its seasonal range (e.g., summering grounds).
 - It will then plateau when the animal is settled in that range.
 - Later in the year, as the animal begins its return migration, the NSD will decrease, eventually approaching zero (or close to zero) if the animal returns to its original starting location.
 - The shape of the curve helps to identify the timing, duration, and extent (distance) of migration events.
2. **Distinguishing Movement Strategies:**
 - **Range Residency (Sedentary behavior):** If an animal is relatively sedentary and stays within a defined home range, the NSD plot will show a rapid increase initially, then quickly plateau and remain relatively flat over time. This indicates the animal is staying within a limited area.
 - **Dispersal:** For dispersing animals (those permanently moving to a new area), the NSD plot will typically show a continuous increase, eventually leveling off at a new, higher plateau, but not returning to the initial displacement.
 - **Nomadism:** Nomadic animals, which constantly move without fixed home ranges or predictable return patterns, might show a more linear or erratic increase in NSD over time.

Why it's useful:

- **Objective Classification:** NSD provides an objective and quantitative way to categorize different movement behaviors (migration, range residency, dispersal, nomadism) without requiring subjective interpretation of maps or visual inspection of paths.
- **Parameter Estimation:** By fitting mathematical models (like sigmoid functions) to the NSD data, researchers can estimate key parameters of movement, such as:

- The timing of migration (start and end dates).
- The duration of migratory phases.
- The distance covered during migration.
- The size of a home range.
- Comparative Studies: It allows for easy comparison of movement patterns between different individuals, populations, or species, as well as how these patterns might change across different years or in response to environmental factors.

In summary, an NSD vs. time of year plot is a powerful tool in movement ecology that graphically represents an animal's cumulative displacement from its starting point over time, helping researchers understand and quantify various movement strategies, especially migration.

Migration mapping

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Migration mapping

Identifying Geographic Migration

To identify individuals that exhibit geographic migration, we will analyse their movement patterns relative to January 1 (or maybe, relative to their winter range?).

1. *Net Squared Displacement (NSD)*: We will calculate the Net Squared Displacement (NSD) for each caribou, which measures the straight-line distance between an individual's starting point and all subsequent locations. The start date for NSD calculation will be set to January 1 (should we use a different date i.e., a time when caribou are presumed to be on their winter range?).
2. *Classifying Migrants and Residents*: By plotting each caribou's NSD values and GPS locations over time, we will classify individuals.
 - o Geographic Migrant: An individual that moves to a spatially distinct (non-overlapping) summer range for a minimum of 30 days.
 - o Vacillating Migrant: A geographic migrant that occupies a distinct summer range but returns to its winter range two or more times during the summer.
 - o Geographic Resident: An individual that remains on the same geographic range year-round.

Timing of Migrations and Seasonal Definitions

For all geographic migrants, the start and end dates of spring and fall migrations will be determined by visually inspecting NSD plots and GPS location data using the BEACONS Movement Mapper app.

- *Spring Migration*: Begins on the date an individual moves away from its winter range and ends on the date it arrives on its summer range.
- *Fall Migration*: Begins on the date an individual leaves its summer range and ends when it returns to its winter range.
- *Seasons for Migrants*: The winter season is the period between the end of fall migration and the start of spring migration. The summer season is the period between the end of spring migration and the start of fall migration.
- *Seasons for Residents*: For geographic residents, seasons will be defined based on the population-level migration timing of the geographic migrants. The summer season for residents will be the time between the 66th percentile of spring end dates and the 33rd percentile of fall start dates. The winter season will be the time between the 66th percentile of fall end dates and the 33rd percentile of spring start dates.

Delineating Seasonal Ranges

Summer and winter ranges for each caribou will be represented by 95% utilization distributions (UDs) calculated using Brownian Bridge Movement Models (BBMMs).

- *BBMM Calculation*: BBMMs will be calculated at a 30 m x 30 m resolution using GPS locations from the defined summer and winter seasons. The smoothing parameter σ_1 will be estimated using a likelihood function, and σ_2 will be set to 30 m.
- *Range Area and Overlap*: The area of each seasonal range will be quantified. We will also calculate the area and percentage of overlap between an individual's summer and winter ranges.

Migration Routes and Stopover Sites

Migration corridors and stopover sites for all migrating caribou will be identified using BBMMs for

the spring and fall migration periods.

- **Migration Corridors:** 95% UD's will be delineated for each migration event. The GPS data used for this will include the locations from the defined migration period plus locations from 24 hours before the start and after the end of the migration.
- **Stopover Sites:** Stopover sites are defined as spatial clusters of GPS locations within a migration route where an animal remains for at least 12 hours. These sites will be identified as the top 10% of the UD from each migration corridor. Adjacent stopover sites with borders less than 300 m apart will be merged. The number of stopover sites used during spring and fall migrations will be quantified for each individual.

Analyzing Altitudinal Variation

To understand vertical movement patterns, we will extract elevation data for each GPS location from a high-resolution Digital Elevation Model (DEM).

- **Seasonal Elevation Use:** We will calculate the median elevation for each individual during the summer and winter seasons.
- **Annual Elevation Patterns:** To visualize annual patterns, we will plot the mean daily elevation for each caribou using a 14-day moving window to smooth daily fluctuations.
- **Classifying Altitudinal Migration:** Based on these plots, individuals will be classified:
 - **Traditional Altitudinal Migration:** Occupying low-elevation winter ranges and high-elevation summer ranges, with a change in mean elevation >250 m between seasons.
 - **Abbreviated Altitudinal Migration:** Occupying high elevations in both winter and summer but descending to lower elevations in spring and fall.
 - **Altitudinal Resident:** An individual that does not exhibit either of the above patterns.

Classifying Overall Migration Strategy

Each caribou will be assigned a final migration strategy by combining its geographic and altitudinal movement classifications.

1. **Geographic Migrants** (summer range overlaps $<20\%$ of winter range) will be sub-classified based on the distance of their migration route:
 - **Long-Distance Migrant:** Migration distance > 20 km.
 - **Short-Distance Migrant:** Migration distance ≤ 20 km.
 - **Vacillating Migrant:** Any geographic migrant that returns to its winter range ≥ 2 times during the summer.
2. **Geographic Residents** will be classified based on their vertical movements as either Traditional Altitudinal Migrants, Abbreviated Altitudinal Migrants, or Altitudinal Residents.
3. An individual that is both a geographic and altitudinal resident will be classified simply as a Resident.

Habitat-Selection Analyses

To evaluate caribou habitat selection, we will conduct analyses at two distinct spatial scales: the home-range scale (third-order selection) using Resource Selection Functions (RSFs) and a finer, movement-based scale using integrated Step-Selection Analyses (iSSAs). This dual approach allows for a comprehensive understanding of selection patterns across both broad landscapes and localized movement paths.

Home-Range Scale: Resource Selection Functions (RSFs)

We will model habitat selection at the home-range scale using a use-versus-availability design with mixed-effects logistic regression. RSFs compare habitat characteristics at locations used by caribou to those at random "available" locations within their home ranges.

1. *Used and Available Locations*: Used locations will be defined as the GPS telemetry locations collected from each of the 25 caribou. For each used location, we will generate 10 random available locations within that individual's home range.
2. *Domain of Availability*: The home range, representing the domain of availability for each caribou, will be estimated using a 95% kernel density estimate (KDE) calculated from its seasonal GPS locations.
3. *Model Specification*: The RSF will take the form of an exponential model, $w(x) = \exp(\beta_1 x_1 + \dots + \beta_n x_n)$, where $w(x)$ is the relative probability of selection and β_i are the selection coefficients for a vector of predictor covariates. To account for individual variation in selection, individual caribou ID will be included as a random effect in all models.
4. *Model Selection*: We will develop a set of candidate models based on biologically relevant hypotheses for caribou habitat selection. Before modeling, we will test for collinearity between predictor variables using a Pearson correlation test and exclude variables where $|r| > 0.7$. We will also test whether to include key variables like elevation as linear or quadratic terms to account for selection of intermediate values. Models will be ranked using Akaike's Information Criterion corrected for small sample sizes (AICc), and the most parsimonious model among those with $\Delta AICc < 2.0$ will be considered the top model.
5. *Model Validation*: The predictive performance of the top-ranked model will be assessed using five-fold cross-validation, where model coefficients are estimated using 80% of the data and used to predict the relative probability of use for the remaining 20%. The correlation between the predicted values and the observed use will be evaluated using Spearman's rank correlation ($\bar{r}_s$).

Fine Scale: Integrated Step-Selection Analyses (iSSAs)

To evaluate fine-scale habitat selection that incorporates movement behavior, we will use iSSAs. This method compares the habitat attributes of observed caribou movements ("used steps") to a set of random alternative steps.

1. *Used and Random Steps*: A "used step" is the straight line connecting two consecutive GPS locations. For each used step, we will generate 10 random steps originating from the same starting point. These random steps will be generated by sampling from each individual caribou's empirical distributions of step lengths and turning angles, using gamma and von Mises distributions, respectively.
2. *Covariates and Model Specification*: We will extract habitat covariates at the endpoint of each used and random step. The iSSA will be implemented using conditional logistic regression, with step included as a stratum. In addition to habitat variables, movement parameters like step length will be included as covariates to simultaneously model habitat selection and movement behavior. As with the RSFs, individual caribou ID will be included as a random effect to account for individual variation.
3. *Model Interpretation*: A global iSSA model containing all hypothesized covariates will be developed. This approach allows for direct comparison of selection patterns at the fine scale with those identified at the home-range scale by the RSFs. Significant selection or avoidance of a habitat covariate will be determined by whether the 95% confidence interval of its coefficient estimate overlaps zero.